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THE EFFECTS OF CENTRIFUGAL AND CENTRIPETAL FORCES ON PRODUCT DEVELOPMENT SPEED AND QUALITY: HOW DOES PROBLEM SOLVING MATTER?

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Data from high-technology firms in Hong Kong were used to investigate whether the outcomes of problem-solving processes (solutions found, problem-solving speed, and solution quality) mediated the effects of centrifugal forces (decentralization, free flow of information, and reach) and centripetal forces (connectedness, temporal pacing, project leader expertise, and superordinate goal) on product development performance (development speed and product quality). Centrifugal and centripetal forces were indirectly related to performance through problem-solving outcomes, although some direct effects of these forces were also found. The results suggest a more complex model of product development than previously envisaged.

In view of the importance of product development as an engine for growth, profitability, and survival for most firms, researchers have focused on understanding the factors affecting successful product development (see Brown and Eisenhardt [1995] for a review). In the significant amount of work produced so far, however, scholars have linked organizational and project team factors directly to product development performance but have left the more fundamental intervening processes virtually unexplored, as Brown and Eisenhardt (1995: 374) noted.

Viewing product development as a problem-solving process (Clark & Fujimoto, 1991; Van de Ven, 1980), Sheremata (2000) postulated two types of organizational forces—centrifugal and centripetal—that influence product development performance (defined as the joint attainment of product quality, development speed, and cost goals) through intervening problem-solving processes. *Centrifugal forces*, which include decentralization, a free flow of information, and “reach,” are “structural elements and processes that increase the quality and quantity of ideas, knowledge, and information” that a project team can access (Sheremata, 2000: 395). These forces push a project team outward, enabling it to access new ideas and information (cf. Brown & Eisenhardt, 1995). In contrast, *centripetal forces*, which include temporal pacing

and connectedness, are “structural elements and processes that integrate dispersed information, knowledge, and ideas into collective action” (Sheremata, 2000: 398). These forces pull a project team together to ensure unified effort among team members.

Sheremata (2000) identified five outcomes of problem-solving processes, three of which are investigated here: solutions found, problem-solving speed, and solution quality. *Solutions found* is the ease with which a project team identifies a large number of alternative problem solutions (Sheremata, 2000: 394). *Problem-solving speed* refers to the degree of speed associated with finding and implementing a solution. *Solution quality* refers to the degree to which a solution enhances new product quality and reflects the extent to which the solution generates fresh thinking in the project team, is creative, and is the right solution for the problem encountered by the team.¹

In spite of its potential significant value, few have subjected Sheremata’s model to empirical testing. We believe that teasing apart the effects of problem-solving processes as mediators and the effects of centrifugal and centripetal forces on the different components of her dependent variable is an important way to advance the literature. The first purpose of the current study was to empirically validate Sheremata’s claim that problem solving matters in the product development process.

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¹ Because of the sensitivity of the respondents to cost data, I could not examine the two other problem-solving processes (cost of problem solving and quality of trade-off decision) (Sheremata, 2000: 394). Similarly, I did not examine development cost.

I examine the effects of the three outcomes of problem-solving processes described above on two dimensions of product development performance: development speed and product quality. *Development speed* is the time elapsed between the conception and definition of a product and its readiness for market (Kessler & Chakrabarti, 1996); *product quality* is the degree to which the product satisfies customer quality and performance requirements (Clark & Fujimoto, 1991). My second purpose was to extend Sheremata's model in two respects. First, I introduce an additional centripetal force, *superordinate goal*, which refers to project team members' acceptance of overall project goals (Pinto, Pinto, & Prescott, 1993). Second, I examine whether the outcomes of problem-solving processes fully or partially mediate the effects of centrifugal and centripetal forces on development speed and product quality, an issue Sheremata did not explore.

Empirical support for a full mediation model would provide supporting evidence for the input-process-output model that appears to underlie Sheremata's model, evidence that has heretofore been elusive (e.g., Ancona & Caldwell, 1992b; Gladstein, 1984; Smith, Smith, Olian, Sims, O'Bannon, & Scully, 1994). Such empirical support would also give credence to a need for an expansion of managers' and researchers' focus on problem-solving processes. It would, for example, suggest that managers might well place more importance on intermediate problem-solving processes in evaluating the progress of product development projects, rewarding team members, and allocating scarce resources. For researchers, the outcomes of prob-

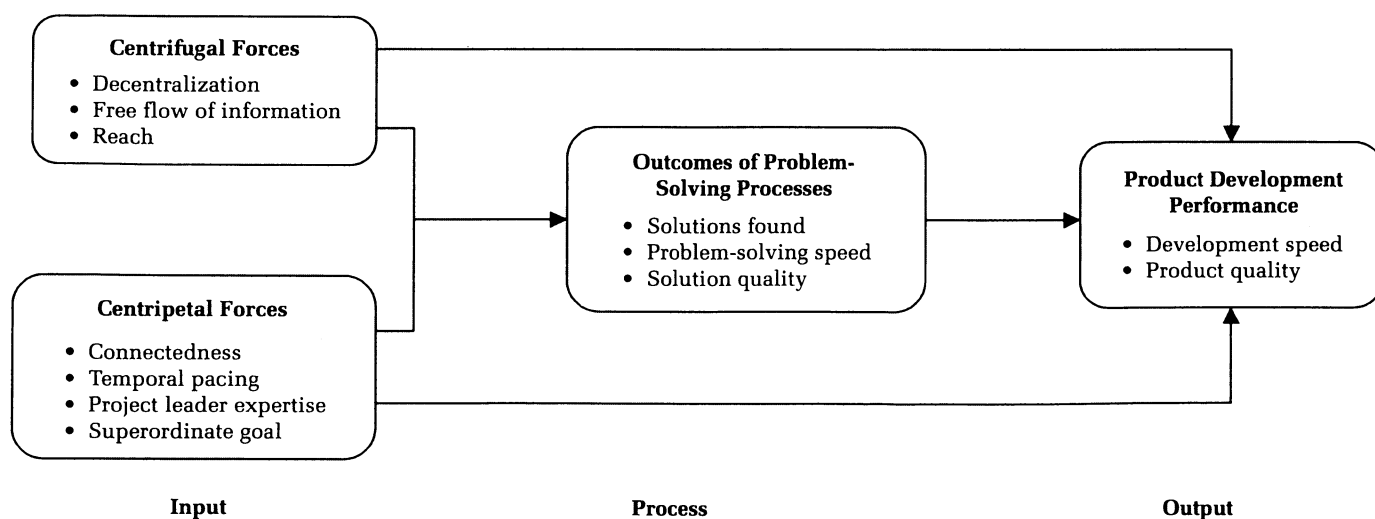
lem-solving processes could become important dependent variables in future research. Support for a partial mediation model would indicate that centrifugal and centripetal forces are more powerful than previously recognized. It would also point to the need for managers to maintain an equal focus on both forces and problem-solving processes.

THEORETICAL BACKGROUND AND HYPOTHESES

Figure 1 presents our theoretical model. Underlying the relationships depicted are the combined theoretical perspectives of the classic input-process-output model of group effectiveness (McGrath, 1984) and the resource-based view of the firm (Barney, 1991). Product development is a disciplined problem-solving process (Brown & Eisenhardt, 1995). It involves searching for ideas and information, processing them, evaluating and selecting from alternative courses of action, and implementing the selected course of action (Sheremata, 2000: 393). Brown and Eisenhardt (1995: 366) distinguished between intermediate product development process goals (such as productivity) and ultimate performance (as assessed by, for instance, product quality). Sheremata (2000) recognized this distinction between intermediate process and ultimate performance goals in product development. She proposed that outcomes of problem-solving processes represent intermediate goals that mediate the effects of centrifugal and centripetal forces on ultimate product development performance.

Although Sheremata did not specifically acknowledge it, the input-process-output model of group

FIGURE 1
Theoretical Model



effectiveness provides theoretical support for her model. According to the input-process-output model, organizational inputs influence group effectiveness entirely through group processes (McGrath, 1984). Although her model may well be consistent with the tenets of the input-process-output model, Sheremata appeared to downplay the power of centrifugal and centripetal forces by not exploring their potential direct effects on product development performance. This omission is problematic, because empirical studies have shown that process variables do not fully mediate the effects of organizational inputs on outputs (e.g., Ancona & Caldwell, 1992; Gladstein, 1984). Here, I extend Sheremata's work by investigating the direct effects of centrifugal and centripetal forces on development speed and product quality.

The resource-based view of the firm (Barney, 1991; Lado & Wilson, 1994) suggests that structure, managerial processes, and interpersonal and intragroup relationships are idiosyncratic capabilities that are valuable, rare, and causally ambiguous to competitors. They are therefore sources of competitive advantage. Centrifugal and centripetal forces clearly fit this description since they represent unique patterns of collecting ideas and knowledge and of integrating resources and structures to harness the diverse skills of people and groups. They are intangible resources that reflect the culture, experience, and history of a particular organization. Not surprisingly, they enable the organization to "create and deliver products and services that are valued by customers" (Lado & Wilson, 1994: 704). For example, studies have found that types of these forces (for instance, team communication, cross-functional interactions, customer and supplier involvement, autonomy, and temporal pacing) directly influence product development performance (see Brown and Eisenhardt [1995] and Kessler and Chakrabarti [1996] for reviews). Brown and Eisenhardt (1995), however, argued that a serious drawback of any direct effects model is that it leaves a mystery about the intervening processes through which organizational factors affect product development performance.

Drawing on the preceding two theoretical perspectives, Figure 1 illustrates the central question of this research: Do problem-solving processes fully or partially mediate the effects of centrifugal and centripetal forces on development speed and product quality? The hypotheses presented below were developed to help in the search for an answer.

Effects of Centrifugal and Centripetal Forces on Problem-Solving Processes

Centrifugal forces. Sheremata (2000) identified three centrifugal forces: decentralization, free flow

of information, and reach. *Decentralization* refers to the delegation of authority to solve problems to lower levels of an organizational hierarchy (Sheremata, 2000: 395). It reflects the autonomy of a project team to make decisions about operational goals and tasks and to solve problems. Such autonomy enhances the likelihood of finding both a large quantity and a high quality of solutions for several reasons. First, team members are closer to the problems that are of concern to them than are those higher in their organizational hierarchies, so the team members are therefore more able to access accurate, high-quality ideas and timely information for solving the problems encountered than are those further away (cf. Burns & Stalker, 1961; Van de Ven, 1980). Second, decentralization increases the quantity and the quality of problem-solving ideas by increasing the cognitive capacity of a team and by reducing the cost of communication and wasteful effort (Clark & Fujimoto, 1991). Third, with decentralization a project team gains a greater sense of control over its work, which increases the urge to seek innovative solutions to problems (Hyatt & Rudy, 1995). However, decentralization may slow down problem solving because of the difficulty of gaining consensus among different people with different sets of ideas for problem solving (Sheremata, 2000).

Free flow of information refers to the extent to which large quantities of rich information can be transferred across individuals and organizational boundaries without encountering resistance (Sheremata, 2000: 397). In the current study, I focus on the access to and transfer of information among individuals in a project team (Burns & Stalker, 1961). Free flow of information facilitates faster generation of a large quantity of ideas and a high quality of ideas by preventing tunnel vision and increasing flexibility and adaptability in problem solving (Allen, 1977; Brown & Eisenhardt, 1995: 368). It also ensures early product definition involving a complete analysis of customer requirements and product features, which increases the likelihood of finding good-quality solutions (cf. Sheremata, 2000). Free flow of information also reduces misunderstanding among team members, which in turn increases the speed and quality of problem solving (cf. Brown & Eisenhardt, 1995; Menon, Jaworski, & Kohli, 1997).

Reach refers to the distance traversed to search for ideas and information (Sheremata, 2000: 396). In this study, I focused on reach to customers, defining this as a project team's interacting with customers to comprehend their needs and market conditions that affect those needs (cf. Sheremata, 2000; Tushman, 1977; Tushman & Nadler, 1978).

Reach increases the availability of a large quantity of good-quality solutions in a timely manner. It does so because reach provides a project team with a larger quantity of new ideas and information and with quicker and clearer understanding of customer and technical problems (Clark & Fujimoto, 1990; Cooper & Kleinschmidt, 1987). Customers can also help to identify and develop the product features necessary to enhance a product's value. Hence, reach increases the chance that a project team will discover good-quality solutions to product design and other technical problems in a timely manner (Clark & Fujimoto, 1990, 1991). Finally, by reaching out to customers a project team gains information and viewpoints that diverge from its own, which should improve the speed and quality of problem solving (cf. Brown & Eisenhardt, 1995). Sheremata (2000) suggested that reach involves significant delays and should therefore slow down problem solving. In the context of reach to customers, research suggests a contrary view, arguing that increased interactions with customers enhance rather than retard speed in product development (see Kessler & Chakrabarti, 1996: 1171). As Van de Ven (1986) argued, direct contact with customers enables quicker action and helps a team uncover solutions to problems. Hence,

Hypothesis 1a. Decentralization is related positively to solutions found.

Hypothesis 1b. Decentralization is related negatively to problem-solving speed.

Hypothesis 1c. Decentralization is related positively to solution quality.

Hypothesis 2a. Free flow of information is related positively to solutions found.

Hypothesis 2b. Free flow of information is related positively to problem-solving speed.

Hypothesis 2c. Free flow of information is related positively to solution quality.

Hypothesis 3a. Reach is related positively to solutions found.

Hypothesis 3b. Reach is related positively to problem-solving speed.

Hypothesis 3c. Reach is related positively to solution quality.

Centripetal forces. Four centrifugal forces were examined here: connectedness, temporal pacing, project leader expertise, and superordinate goal. *Connectedness* refers to direct contact among individuals (Sheremata, 2000: 398) such as project team members, members of other projects, and members

of other organizations. In the current study, I defined connectedness more narrowly as the degree of interaction or direct contact between a project team and other organizational functions (Ancona & Caldwell, 1992; Clark & Fujimoto, 1990). Connectedness leads to finding solutions, fast problem solving, and good-quality solutions for three reasons. First, it allows the integration of ideas in problem solving by ensuring the availability of the expertise and resources of other functional groups to a project team. Second, connectedness strengthens the links between a project team and other functions, ensuring greater opportunities for feedback to the project team (cf. Clark & Fujimoto, 1990). Third, connectedness ensures the support of other functions for a project team's activities. Such support allows a team to tap into a firmwide knowledge base for new ideas and information to integrate with its own in problem solving.

Temporal pacing refers to the use of a formal procedure with clear guidelines, rules, and schedules for organizing, carrying out, and assessing the progress of a project team's activities (Gersick, 1994; Griffin, 1997). Temporal pacing enhances the likelihood of finding solutions and problem-solving speed by infusing a project team with a sense of urgency and awareness of a need to solve problems early to meet set deadlines (Gersick, 1994). It also facilitates effective senior management monitoring of a project (cf. Brown & Eisenhardt, 1995). Such monitoring motivates team members to find new ideas to solve problems because it signals them and others in their firm that the project is important (Griffin, 1997). Indeed, without milestones there is likely to be a sense of disorder that may hamper early identification of ideas to solve problems (Thomke & Fujimoto, 2000). Sheremata (2000) contends that temporal pacing may reduce solution quality because its overriding focus is on speed. However, I argue that temporal pacing enhances solution quality by providing a sense of structure, a common frame of reference for a team's members, and periodic feedback about the quality and completeness of the team's activities (Cooper & Kleinschmidt, 1994). Indeed, research suggests that temporal pacing reduces wasteful explorations and errors in problem solving. For example, temporal pacing allows easier surfacing of problems. It also ensures that diverse ideas from team members are integrated at specific development stages to solve those problems. This improves the quality of the problem-solving activities (McDonough & Barczak, 1991; Meyers & Marquis, 1969).

The last two centripetal forces examined are project leader expertise and superordinate goal, the latter being my addition to Sheremata's model.

Although both forces may be sources of ideas and information, I believe that they integrate more than they increase access to ideas and information. *Project leader expertise* refers to the degree to which a project's leader possesses skills, knowledge, and experience relevant to both technical and management aspects of the project (Nord & Tucker, 1987; Sheremata, 2000). This construct is used in place of project leader influence since it is the main source of such influence (Sheremata, 2000). An expert project leader is an integrator because he or she is able to motivate a team for collective action. He or she increases the likelihood of finding good-quality solutions through an ability to motivate the project team to combine quality ideas into new knowledge, to monitor the team's progress, and to provide corrective feedback (Sheremata, 2000: 399). Problem solving is dependent on the actions of different team members and others outside it. An expert project leader is able to reduce problem-solving time because of an ability to coordinate and solve problems among and between team members and other functional groups (cf. Clark & Fujimoto, 1990, 1991).

Superordinate goal refers to the degree to which project team members accept and identify with the overall goals of a project (Pinto et al., 1993). Problem solving is not achieved by merely putting people and structures together but also requires offering them a shared goal for collective achievement. As Dougherty pointed out, "It is desirable that people feel free to generate ideas, create possible solutions to problems, and experiment with various courses of action. It is also desirable that people feel responsible to work hard towards common goals" (1996: 430). Consequently, superordinate goal serves as a useful proxy for project team influence. This is because by sharing a common goal, project team members can use their collective influence to obtain and use resources to facilitate problem solving. A superordinate goal enhances the likelihood of finding good-quality solutions in a timely manner because team members with a common goal become more open to the diverse perspectives of each other and effective in integrating a larger pool of good-quality ideas and information (Hyatt & Rudy, 1997). It speeds up problem solving because the feeling of psychological ownership among team members facilitates communication and reduces misunderstanding in problem solving, thereby preventing wasteful effort (cf. Pinto et al., 1993). This discussion suggests the following hypotheses:

Hypothesis 4a. Connectedness is related positively to solutions found.

Hypothesis 4b. Connectedness is related positively to problem-solving speed.

Hypothesis 4c. Connectedness is related positively to solution quality.

Hypothesis 5a. Temporal pacing is related positively to solutions found.

Hypothesis 5b. Temporal pacing is related positively to problem-solving speed.

Hypothesis 5c. Temporal pacing is related positively to solution quality.

Hypothesis 6a. Project leader expertise is related positively to solutions found.

Hypothesis 6b. Project leader expertise is related positively to problem-solving speed.

Hypothesis 6c. Project leader expertise is related positively to solution quality.

Hypothesis 7a. Superordinate goal is related positively to solutions found.

Hypothesis 7b. Superordinate goal is related positively to problem-solving speed.

Hypothesis 7c. Superordinate goal is related positively to solution quality.

Effects of Problem-Solving Processes on Development Speed and Product Quality

The outcomes of problem-solving processes constitute the basic ingredients of product development performance (Sheremata, 2000; Van de Ven, 1980, 1986). The link is based on the resource-based view of the firm (Barney, 1991). Per this view, problem-solving processes are intangible, higher-order capabilities that enable a firm to perform critical product development activities better than its competition. For example, by ensuring testing and solving problems, problem-solving processes may promote faster learning and competence development in a project team. This learning may enhance the development of knowledge of customer and operational requirements. Further, these problem-solving processes make it likely that a product will contain ideas about market and technology that may be more advanced than what competing products have. Thus, project teams able to find a large amount of solutions faster are more likely to overcome operational and technical problems in a timely manner, thereby speeding up the development process and product quality (Clark & Fujimoto, 1990, 1991; Sheremata, 2000). Similarly, high-quality solutions allow a project team to solve problems associated with the performance require-

ments of customers such as reliability and durability, and to do so in a timely manner. It follows that high-quality solutions increase both development speed and product quality (cf. Brown & Eisenhardt, 1995). Formally, I posit that:

Hypothesis 8a. Outcomes of problem-solving processes (solutions found, problem-solving speed, and solution quality) are related positively to development speed.

Hypothesis 8b. Outcomes of problem-solving processes (solutions found, problem-solving speed, and solution quality) are related positively to product quality.

Partial Mediating Effects of Problem-Solving Processes

The hypotheses presented above link centrifugal and centripetal forces to three outcomes of problem-solving processes, and these in turn are linked to development speed and product quality. Implicitly, the discussion suggests that all the effects of these forces on development speed and product quality are worked through the problem-solving processes, suggesting full mediation. However, as mentioned earlier, such a proposition downplays the power of centrifugal and centripetal forces. It is also inconsistent with prior research showing that several types of these forces have direct effects on product development performance (see Brown & Eisenhardt, 1995). Further, studies of group effectiveness indicate that group processes do not fully mediate the effects of organizational inputs on group performance (Gladstein, 1984). For example, Ancona and Caldwell (1992) examined how internal team processes (such as setting goals and priorities) intervene between group diversity and performance and found that the process variables did not fully explain the observed effects of group diversity on performance. Similarly, Smith and colleagues (1994) found that process variables did not mediate the effects of top management team demography and organizational factors on performance. Finally, Pinto and coauthors (1993), in a study of cross-functional teams, showed that organizational factors like superordinate goals had effects that went beyond the effects of the process variable (cross-functional cooperation) to the ultimate project goals. In brief, the literature shows that organizational factors have both direct and indirect effects on group performance through process variables. As mentioned earlier, Sheremata did not explore this possibility. I extended her work by positing:

Hypothesis 9a. The effects of centrifugal and centripetal forces on development speed are partially mediated by outcomes of problem-solving processes (solutions found, problem-solving speed, and solution quality).

Hypothesis 9b. The effects of centrifugal and centripetal forces on new product quality are partially mediated by outcomes of problem-solving processes (solutions found, problem-solving speed, and solution quality).

METHODS

Sample

The sample for the study contained 300 Hong Kong manufacturing firms in the computer, electronics, information technology, and software industries randomly selected from the 1999 directory of Hong Kong firms published by Kompas (Hong Kong) Ltd. I contacted each firm by phone to inquire if it had an ongoing product development project involving multiple functions and if its top manager or managers would be willing to provide access to different individuals involved in the project at different times. Of the 120 firms whose representatives agreed to participate, 104 firms provided responses, for an effective participation rate of 35 percent. To assess the representativeness of the sample, I compared participating firms with a sample of nonparticipating firms on firm age and number of employees, using a *t*-test. No significant differences were found.

The average age of firms in the sample was 11.95 years; the average number of employees was 410.32; average project team size was 9.40 individuals; and the average project duration was 8.76 months. Eighty-four percent of the projects were developing incremental products (defined as improvements and adjustments in current technology and products), and the rest were developing radical products (defined as fundamental and revolutionary changes in technology or products) (Dewar & Dutton, 1986). Although innovations can be described along a continuum of level of new knowledge, we followed Dewar and Dutton (1986: 1423) in focusing on the polar types because middle values are difficult to interpret. Forty-six percent of the projects that comprised the sample for this study were developing software products, and the remainder concerned hardware products. Finally, 80 percent of the participating firms were under time pressure to develop the product of interest here.

Measures and Validation

Given the prevalence of consensus decision making among the Chinese and the varied natures of centrifugal and centripetal forces, I asked two team members in each participating project to provide a single consensual rating for the measures of these forces. I selected the team members upon the advice of senior managers who deemed them as being highly knowledgeable about their projects. The team leader of each project provided the data for the measures of the outcomes of problem-solving processes and the project characteristics a few weeks later. The data on development speed and product quality were obtained from the senior manager or CEO who had overall responsibility for the project at its completion. Prior to the study, I took great care to evaluate the instrument, which was drawn from various sources written in English, because of the potential language barrier in Hong Kong. I therefore conducted 15 in-depth interviews with project managers to ensure the clarity and relevance of the constructs and their measures. No problems were encountered in administering the English-language questionnaire on-site to respondents. The measures and their sources are presented in the Appendix. Their construct reliabilities, all but two of which are above the commonly accepted threshold of .70, are in Table 1.

I performed three separate confirmatory factor analyses (CFAs), grouping problem-solving processes, development speed, and product quality into the first model. Centrifugal and centripetal forces formed the second and third confirmatory factor analysis models, respectively. The Appendix presents the results of the CFAs. The models fitted the data reasonably well. Table 1 presents the correlations, descriptive statistics, and reliabilities of the constructs.

ANALYSIS AND RESULTS

Following prior research (e.g., Pinto et al., 1993; Smith et al., 1994), I tested the hypotheses with regression analysis, using procedures appropriate for assessing interdependent variables, since problem-solving processes are interdependent. Smith and colleagues (1994), in their study of demography, group process, and performance, assessed the effect of one intervening process variable (informality of communication) on another (social integration). They treated social integration as an "intervening dependent variable" while "all team demography and control variables, plus informality of communication were treated as independent variables" (1994: 429). I controlled for team size,

time pressure (whether or not a team felt time pressure in developing the product; 1 = "yes," 0 = "no"), type of product (1 = "radical," 0 = "incremental"), firm size (number of employees), and the nature of the product (0 = "software," 1 = "hardware"). None of this study's variables had a variance inflation factor above 2.0, indicating that multicollinearity was not a significant problem. Table 2 presents the results of the regression analyses.

Consistent with Hypothesis 1c, the findings presented in Table 2 show that decentralization was significantly related to solution quality. Free flow of information was related to problem-solving speed (Hypothesis 2b). The results suggest that reach was related positively to solutions found and to problem-solving speed, in support of Hypotheses 3a and 3b, respectively. No other hypothesized effects of centrifugal forces were supported. With respect to centripetal forces, the results show that temporal pacing was significantly related to problem-solving speed and solution quality, supporting Hypotheses 5b and 5c. Consistent with Hypothesis 7c, superordinate goal was related positively to solution quality. No other hypothesis relating to centripetal forces was supported. The data in model 4 in Table 2 show partial support for Hypothesis 8a. Solutions found and problem-solving speed were significantly related to development speed. Problem-solving speed and solution quality were related significantly to product quality, in partial support of Hypothesis 8b (model 5).

To test the partial mediation hypotheses (9a and 9b), I conducted a three-stage regression analysis to determine whether the three conditions for mediation (Baron & Kenny, 1986) were satisfied. The data in Table 2 (models 4 and 5) permit a test of the first condition for mediation, that the proposed mediators significantly predict the dependent variables. Models 1, 2, and 3 permit a test of the second condition, prediction of the mediators by the independent variables. Table 3 presents the results of hierarchical regressions testing the third condition for mediation. This condition requires the contribution of the independent variables to drop substantially for partial mediation and to become insignificant for full mediation when they are entered into the model together with the mediators.

The outcomes of problem-solving processes add 6 percent to the explained variance in development speed (model 3). One outcome, problem-solving speed, was significantly related to development speed. However, decentralization, reach, and project leader expertise also remained significant. The outcomes of problem-solving processes add 9 percent to the explained variance in product quality (model 6). Two outcomes, prob-

TABLE 1
Correlation Matrix and Descriptive Statistics^a

Variable	Mean	s.d.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1. Product quality	3.68	0.70	.87															
2. Development speed	3.23	0.91	.47**	.91														
3. Solutions found	3.08	0.59	.29**	.31**	.56													
4. Problem-solving speed	3.14	0.87	.53**	.46**	.28**	.86												
5. Solution quality	3.39	0.71	.48**	.29**	.27**	.31**	.76											
6. Decentralization	3.32	0.72	.14	-.13	.01	.17	-.10	.76										
7. Free flow of information	3.70	0.69	.25**	.22**	.13	.53**	.33**	.22**	.87									
8. Reach	3.27	0.75	.36**	.45**	.42**	.33**	.36**	-.00	.23**	.79								
9. Connectedness	3.26	0.59	.10	.04	.17	.08	.23**	.03	.30**	.32**	.64							
10. Temporal pacing	3.43	0.67	.55**	.31**	.24**	.44**	.54**	.07	.24**	.25**	.12	.80						
11. Project leader expertise	3.40	0.73	.49**	.52**	.28**	.35**	.34**	-.07	.27**	.42**	.16	.37**	.78					
12. Superordinate goal	3.51	0.63	.38**	.27**	.15	.37**	.42**	.25**	.67**	.36**	.34**	.28**	.35**	.87				
13. Project team size			.11	-.11	.09	-.14	.02	.05	-.11	.17	.22**	.07	-.07	.06				
14. Time pressure	9.40	9.02	-.15	-.14	-.01	.19**	-.28**	.14	-.05	-.06	-.20**	-.28**	-.23**	-.10	-.07			
15. Type of product			.08	.01	.07	.16	.12	.10	.03	-.16	.04	.10	.05	.09	-.09	.13		
16. Firm size	410.32	980.29	.14	-.01	-.09	.16	.11	.16	.04	.03	.02	.29**	.08	.20**	.27**	-.13	.03	
17. Software vs. hardware			-.06	.06	.09	.02	.03	-.10	.03	.12	.14	.05	.22**	.05	.00	.01	-.01	.02

^a Coefficient reliabilities are on the diagonal in bold.

** $p < .01$

*** $p < .001$

TABLE 2
Results of Regression Analyses for Three Problem-Solving Processes as Dependent Variables^a

	Model 1	Model 2	Model 3	Model 4	Model 5
Variables	Solutions Found	Problem-Solving Speed	Solution Quality	Development Speed	Product Quality
Control					
Project team size	.11	-.14*	-.01	-.07	.14*
Time pressure	.10	-.15*	-.11	-.04	.00
Radical vs. incremental product	-.02	.09	.06	-.06	-.02
Firm size	-.19*	.12	-.02	-.06	.03
Software vs. hardware products	.03	-.01	-.06	.04	-.09
Outcomes of problem-solving processes					
Solutions found		.15*	.08	.17*	.07
Problem-solving speed	.21*		-.07	.36***	.45***
Solution quality	.11	-.06		.14	.32***
Centrifugal forces					
Decentralization	.03	-.08	.19**		
Free flow of information	-.10	.48***	.10		
Reach	.25***	.19**	.12		
Centripetal forces					
Connectedness	.05	-.14	.01		
Temporal pacing	.09	.22**	.41***		
Project leader expertise	.09	.04	.00		
Superordinate goal	-.04	-.10	.23**		
<i>R</i> ²	.27	.49	.46	.27	.41
Adjusted <i>R</i> ²	.15	.42	.37	.21	.36
<i>F</i>	2.37**	6.28***	5.36***	4.49***	8.46***
<i>n</i>	103	103	103	103	103

^a Standardized regression coefficients are reported.

* $p < .05$

** $p < .01$

*** $p < .001$

lem-solving speed and solution quality, were significantly related to product quality. Nevertheless, free flow of information, temporal pacing, project leader expertise, and superordinate goal all remained significant predictors of product quality. These results indicate that the outcomes of problem-solving processes partially mediate the effects of centrifugal and centripetal forces on development speed and product quality. This evidence supports Hypotheses 9a and 9b both. Attesting to the interdependence of the outcomes of problem-solving processes (Sheremata, 2000), solution found and problem-solving speed predicted each other. However, we found no relationships between these two outcomes and solution quality.

DISCUSSION AND IMPLICATIONS

This study provides one of the first empirical tests of the intervening role of the outcomes of problem-solving processes (solutions found, prob-

lem-solving speed, and solution quality) in the relationship between centrifugal and centripetal forces, and product development performance (development speed and product quality). In support of our expectations, the effects of the two sets of forces on development speed and product quality were not fully, but rather, were partially mediated by the outcomes of problem-solving processes. These results extend Sheremata's (2000) model, showing that the relationships are more complex than previously envisaged.

Implications for Theory

The theoretical conclusions of this research are threefold. First, examining the direct effects of centrifugal and centripetal forces on development speed and product quality leads to the conclusion that they are sources of competitive advantage. The results are consistent with resource-based theory, suggesting that these forces are knowledge-based, firm-specific, and socially complex capabilities (cf.

TABLE 3
Results of Hierarchical Regression Analysis of the Mediating Role of Outcomes of Problem-Solving Processes^a

Variables	Development Speed			Product Quality		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Control						
Project team size	-.11	-.10	-.07	.08	.10*	.16*
Time pressure	-.15	-.02	-.01	-.14	.02	.08
Radical vs. incremental product	.02	-.06	-.08	.09	.00	-.03
Firm size	-.01	-.06	-.08	.08	-.06	-.09
Software vs. hardware	.07	-.04	-.04	-.06	-.15*	-.13*
Centrifugal forces						
Decentralization		-.13	-.16*		.09	.09
Free flow of information		.02	-.12		.15*	.22**
Reach		.31***	.23**		.10	.01
Centripetal forces						
Connectedness		-.13	-.09		-.07	-.03
Temporal pacing		.14	.07		.39***	.22**
Project leader expertise		.30**	.28***		.32***	.30***
Superordinate goal		.09	.14		.17*	.17*
Outcomes of problem-solving processes						
Solutions found			.05			-.00
Problem-solving speed			.30***			.37***
Solution quality			-.12			.20**
Incremental R^2		.36	.06		.41	.09
Partial F		7.77***	2.95**		10.30***	6.08**
R^2	.04	.40	.46	.05	.47	.56
Adjusted R^2	.00	.32	.36	.01	.41	.49
F	.83	5.05***	4.89***	1.19	6.87***	7.64***
n	103	99	91	103	98	91

^a Standardized regression coefficients are reported.

* $p < .05$

** $p < .01$

*** $p < .001$

Barney, 1991; Lado & Wilson, 1994). The results indicate that decentralization is a double-edged sword, having a negative effect on development speed but a positive effect on solution quality. Free flow of information enhances problem-solving speed and product quality. This finding attests to its capacity to improve the timely availability of high-quality new ideas (Sheremata, 2000). As a project leader in the field put it, "We missed our launch date by nearly two years due to design and customer problems largely because some team members used information as a power tool and were reluctant to share information in a timely manner." The results indicate that reach is related positively to solutions found and problem-solving speed but not to solution quality or product quality. Recall that our measure of reach focused narrowly on customers and did not cover reaching out to R&D specialists, suppliers, and other sources of information (see Sheremata, 2000: 396). Hence, a

plausible explanation is that reaching out to customers, without reaching out to sources such as R&D specialists or suppliers, may bias problem solving toward solutions that are of lower quality and technically inferior. This observation reinforces the notion that customers are unable to articulate their needs and that firms reaching too close to customers may miss out on emerging technology opportunities and related insights that could provide radical solutions to problems in product development (cf. Christensen & Bower, 1996).

Regarding centripetal forces, we found a positive relationship between temporal pacing and problem-solving speed, solution quality, and product quality. The latter finding refutes Sheremata's (2000) argument that temporal pacing tends to enhance problem-solving speed at the expense of solution quality. Surprisingly, the evidence suggests that project leader expertise is a direct predictor of

development speed and product quality but not of the outcomes of the intermediate problem-solving process. Finally, superordinate goal predicted solution quality and product quality, suggesting that it helps to promote teamwork to facilitate quality in both the intermediate processes and the ultimate performance of a project.

Second, this research sheds light on the important intervening role of problem-solving processes in product development. The results indicate that although centrifugal and centripetal forces affect the outcomes of problem-solving processes, these outcomes influence product development performance. For example, problem-solving speed is related positively to development speed. Both problem-solving speed and solution quality are positively related to product quality. These results suggest that centrifugal and centripetal forces have even greater power than previously thought. These forces have spillover effects on development speed and product quality that go beyond their effects on the outcomes of problem-solving processes. These indirect effects of centrifugal and centripetal forces lend some support to Sheremata's (2000) theory that problem-solving processes mediate the effects of these forces on product development performance. However, in an important extension of Sheremata's model, I show that the outcomes of problem-solving processes play a partial rather than a full mediating role. This finding receives support from research suggesting that the input-process-output model, which appears to underlie Sheremata's model, oversimplifies the relationships between organizational inputs, processes, and outputs (e.g., Gladstein, 1984; Pinto et al., 1993; Smith et al., 1994). Note, however, that Sheremata examined the joint attainment of development speed, product quality, and cost goals. The extent to which the present conclusion about the input-process-output model applies to such a composite dependent variable remains an issue for future research to address. In summary, the results of this study point out the complexity of product development and the need to separate out the effects of different outcomes of the intervening problem-solving processes on different dimensions of ultimate product development performance.

Finally, this research indicates that although Sheremata's (2000) model focuses on radical innovations, it appears to reflect both incremental and radical product development, given that most of the projects in my sample involved incremental products. The results obtained here imply that although radical product development is undoubtedly more complex than incremental product development, the latter still remains a complex task.

Hence, scholars need to pay attention to the intermediate problem-solving processes and ultimate performance goals in incremental product development. This study contributes reliable and valid measures for Sheremata's constructs that should prove useful tools for future research.

Managerial Implications

The first implication of these results for managers is that centrifugal and centripetal forces need attention, not only because they have effects on the outcomes of problem-solving processes, but also because they have direct effects on development speed and product quality. Second, problem-solving processes do matter because their outcomes may act as conduits, albeit partially, allowing centrifugal and centripetal forces to affect development speed and product quality. Therefore, project managers should consider the effects of these forces both on the intervening problem-solving processes and on ultimate product development performance. To do this, project managers should clearly delineate the outcomes of intermediate problem-solving processes and ultimate product development performance goals. Although product development is a problem-solving process (Brown & Eisenhardt, 1995), few managers promote and reward achievements in important intermediary processes and attain the kind of important intermediate outcomes they need to attain ultimate product development performance goals. My results suggest that project managers unable or unwilling to consider these differences in their resource allocation and reward strategies for project teams may be at a competitive disadvantage.

Limitations and Future Research

The study reported here has some limitations. First, the generalizability of the findings is limited, since my sample came from a particular context, where, it can be argued, product development is embryonic. Second, the study is only a partial test of Sheremata's model, since I did not consider two other outcomes of problem-solving processes (the cost of problem solving and the quality of trade-off decisions) and ultimate development cost. Sheremata also proposed that the outcomes of problem-solving processes combine to influence the joint attainment of product quality, development speed, and cost goals. This part of her model remains to be empirically tested. Another limitation of my study is that I defined constructs such as reach and connectedness more narrowly than Sheremata did. In future research, taking a broader perspective on

these constructs than I did here may provide greater insight.

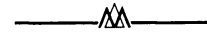
The measures used here are subjective; therefore, I cannot completely rule out the effects of the shortcomings associated with subjectivity. However, this measurement approach was deemed appropriate here. The magnitude of the problems encountered in product development, the number of people solving them, and their inherent difficulties all differ markedly from project to project. Hence, it is inherently difficult to measure constructs like problem-solving speed and solution quality with objective measures and to compare findings across projects. However, I collected the data from different informants to reduce the potential for causal attributions between the variables (Ayers, Dahlstrom, & Skinner, 1997). The reliabilities of two of the constructs (solutions found and connectedness) were below the widely accepted threshold of .70. This should suggest cautious interpretation of the associated findings. Finally, target dates and quality objectives change during product development. Hence, it does matter whether development speed or product quality reflects initial planned targets or targets set near the completion of a project. Unfortunately, the current measures do not reflect such changes in targets, a fact that should be taken into account in interpreting the findings.

In spite of these limitations, I believe that this study adds value to the literature. It emphasizes that problem-solving processes are important to the attainment of product development performance. However, it also suggests that centrifugal and centripetal forces play an important role directly and indirectly through problem-solving processes, in enhancing product development performance. By empirically exploring the notion of partial mediation of the effects of these forces through problem-solving processes, I have opened up a fruitful avenue for future research on product development. I hope the new measures developed and validated in this study will prove useful to researchers in future research endeavors.

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This study continues with an appendix on the next page.

APPENDIX

Measures and Confirmatory Factor Analysis Results

Construct	Operational Measure ^a	Factor Loading	<i>t</i>
Model 1 $\chi^2/df = 1.51$, RMSEA = .06, GFI = .86, CFI = .95			
Development speed ^b	How does each of the following statements describe the product development project?		
	The project duration met the planned time schedule.	.79	8.50
	The project was completed on time as planned.	.80	8.31
	The length of the project met the planned objectives better than previous projects.	.82	8.52
	The speed of development was faster than previous projects.	.93	10.10
Product quality ^c	How does each of the following statements describe the product development project?		
	The product conformed to performance specifications required by customers.	.84	10.05
	The product performance met the requirements of customers.	.87	10.86
	The design of the product met the quality standards expected by the firm's customers.	.79	9.20
Solutions found ^b	Every product development product project faces technical, operational and other problems. Focusing on problems that your project team encountered during the project, to what extent do you agree or disagree with the following statements?		
	We identified several alternative solutions for each problem the project team encountered.	.75	6.85
	The project team seemed to always come up with solutions to problems encountered.	.73	6.70
	It was always easy to identify new solutions to the project team problems.	.41	4.01
Problem-solving speed ^b	Every product development product project faces technical, operational and other problems. Focusing on problems that your project team encountered during the project, to what extent do you agree or disagree with the following statements?		
	Solutions found for problems we faced were not timely. (reverse-coded)	.89	10.89
	The project team was very slow in finding and implementing solutions to the problems we encountered. (reverse-coded)	.85	10.23
	Ideas for solving the problems encountered were discovered rather late to be implemented successfully. (reverse-coded)	.76	8.63
Solution quality ^b	Every product development product project faces technical, operational and other problems. Focusing on problems that your project team encountered during the project, to what extent do you agree or disagree with the following statements?		
	The solutions found generated fresh thinking about product quality in the project team.	.54	5.12
	Solutions found for the problems encountered were very creative.	.56	5.35
	The solutions found were quality solutions for the problems we faced.	.80	7.45
	The solutions found were exactly what we needed. ^d		
Model 2 $\chi^2/df = 1.82$, RMSEA = .08, GFI = .85, CFI = .90			
Centrifugal forces	Indicate your agreement with these statements regarding the project team's product development activities.		
Decentralization ^e	We had to ask a higher manager before we could do almost anything. (reverse-coded)	.68	7.22
	Even small matters had to be referred to someone higher up for approval. (reverse-coded)	.90	10.15
	There could be little action taken in the project until a higher manager approved. (reverse-coded)	.49	4.88
	Any decision we made had to have a higher manager's approval. (reverse-coded)	.36	3.44
	A team member who wanted to make his/her own decisions would be quickly discouraged. (reverse-coded)	.67	7.11

APPENDIX Continued

Construct	Operational Measure ^a	Factor Loading	<i>t</i>
Free flow of information ^f	The project team members did not hide any issues from each other.	.71	7.88
	Channels of communication among team members were open.	.87	10.51
	Access to information from team members was quick and easy.	.84	10.02
	Project team members engaged in open and honest communication.	.66	7.14
	Team members willingly kept each other informed at all times.	.66	7.09
Reach ^b	Team members put a lot of effort in interacting with potential customers.	.83	9.11
	Team members had frequent interactions with likely customers of the product.	.81	8.93
	The project team members collected information about new market developments affecting customers.	.62	6.45
	We monitored and learnt about new technology developments affecting customer requirements.	.52	5.19
Model 3 $\chi^2/df = 1.27$, RMSEA = .05, GFI = .88, CFI = .95			
Centripetal forces	Indicate your agreement with these statements regarding the project team's product development activities.		
Connectedness ^c	The project team had close, personal interactions with other members of the firm regarding the project.	.73	5.96
	We had frequent consultation with other departments in the firm regarding the project.	.59	5.11
	The project team put a lot of effort into building relationships with other functions regarding the project.	.55	4.81
Temporal pacing ^b	The project used a formal process with frequent reviews of progress.	.68	7.17
	The project used a formal process with different stages for major activities.	.91	9.18
	The progress towards the project objectives was reviewed at specific stages.	.84	8.10
	The project team followed a documented process with specific milestones for each activity.	.58	6.07
Project leader expertise ^b	The project leader had experience in both technical and management aspects of product development projects.	.82	7.83
	The project leader had diverse technical and management knowledge and experience in the industry.	.69	6.70
	The project leader had significant technical and management skills and knowledge about the team's project.	.55	5.30
Superordinate goal ^f	Team members were all committed to the same project goals.	.87	10.56
	The project goals and objectives linked all of us together.	.83	9.84
	Team members accepted the project goals as their own.	.73	8.14
	Every team member behaved in ways that supported the overall goals of the project.	.70	7.66
	The project had specific objectives and each team member was concerned as if they were personal goals.	.69	7.63

^a The senior manager with overall responsibility for each product development project completed two measures: development speed and product quality. The team leader of each project completed three measures: solutions found, problem-solving speed, and solution quality. The remaining measures were based on single consensus ratings completed by two project team members. The response format for all items was 1, "strongly disagree," to 5, "strongly agree."

^b New scale developed for this research.

^c Scale originated by Menon, Jaworski, and Kohli (1997).

^d Item was deleted.

^e Scale was adapted from Ayers, Dahlstrom, and Skinner (1997).

^f Scale was adapted from Hyatt and Rudy (1997).